

Unaugmented 3D Navier-Stokes: the proof chain

Track B - WRITE (6) open - not QED

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Abstract

This is the written chain for classical three-dimensional incompressible Navier-Stokes: no Q1, keep $1/r^4$. Lines (1)-(5) sit. Line (6) does not. Lines (7)-(9) wait on (6). Emitting this paper is not a close. Theorem A (the Q1-augmented PDE) is a different equation and is not used here. Check B stays open.

1. Aimed theorem (does not sit)

Let u be a smooth solution of three-dimensional incompressible Navier-Stokes (periodic or whole space), viscosity $\nu > 0$, no Q1, keep $1/r^4$. Let $X = \|\omega\|_2^2$. Then X stays finite on $[0, T]$ for arbitrary T , and u remains smooth.

That statement is the leftover. It is not a theorem of this desk.

2. The leftover form

$$\frac{d}{dt} X + \nu \|\text{grad } \omega\|_2^2 \leq \epsilon \nu \|\text{grad } \omega\|_2^2 + C_\epsilon X R(t), \quad \int_0^T R < \infty.$$

The only term that can beat viscosity is the stretching leftover. F is not this object. The extra Q1 stress is not on this equation.

3. Proof chain

(1) HAVE - Energy.

Leray: $\int_0^T X(t) dt < \infty$ on these packets, and the energy inequality holds.

(2) HAVE - Enstrophy identity.

$$\frac{d}{dt} X + \nu \|\text{grad } \omega\|_2^2 = - \int \omega \cdot S \omega$$

up to lower-order terms already controlled.

(3) HAVE - Leftover form.

Absorb a slice of dissipation to obtain the display in section 2. The stretching leftover is the only term that can beat viscosity.

(4) HAVE - Split.

$\int \omega \cdot S \omega = \text{hole 1 (aligned } P^+ \text{ on } E_c) + \text{hole 2 (unaligned } P^+ \text{ on } E_c) + \text{hole 3 (off } E_c)$.
Scored on the $n=32$ box as B37.

(5) HAVE - Named blanks.

A_1 = alignment in time for all data (hole 1). $A_2 = \int \|\lambda_2^+\|$ for all data (live cubic; Miller cut B38). On this box A_1 is off and A_2 is live and did not blow on the B15 path (B40, B41). Neither integral is known for all data.

(6) WRITE - the leftover. NOT DONE.

One all-data integrable residual: $\int_0^T R(t) dt < \infty$, or all-data A1, or all-data A2, or a field that kills the stretching leftover.

(7) THEN - Gronwall. Waits on (6).

From (3) and (6), $X(t)$ stays finite on $[0, T]$.

(8) THEN - Continuation. Waits on (6).

Beale-Kato-Majda: if $\int_0^T \|\omega\|_{\infty} dt < \infty$ then the solution continues. L^2 is not the max.

(9) THEN - Bootstrap. Waits on (6).

A bound on X and no blowup of $\|\omega\|_{\infty}$ upgrades to smoothness on $[0, T]$. If T is arbitrary, the solution is globally regular.

If (6) sits, (7)-(9) close the aimed theorem. (6) does not sit.

4. Completion ledger

(1)-(5) done.

(6) all-data $R / A1 / A2 /$ killing field - not done.

(7)-(9) Gronwall / continuation / bootstrap - waiting on (6).

Printing this paper does not change that table. Please-finish-bad does not change that table. Theorem A does not change that table.

5. Candidates for (6)

Classify one:

- all-data alignment in time (A1)
- all-data integral $\|\lambda_2^+\|$ (A2)
- a different integrable residual
- a killing field for the stretching leftover

Do not use Q1, Phi-cancel, a Q-floor, or leftover-close B42. Do not spawn $n=64$. Do not cash an $n=32$ reading as an a priori.